

Algebra 1

Grant Talbert

November 18, 2025



MA741

Contents

1	Categories	2
1.1	Definition	2
2	Groups	4

Lecture 1: Syllabus day

Tue 02 Sep 2025 15:29

Syllabus at <https://juanitaduquer.github.io/Documents/syllabus741.pdf>.

Chapter 1

Categories

1.1 Definition

Definition 1.1.1. A category $\mathcal{C}$ has

- a class $\text{Obj}(\mathcal{C})$ of *objects*;
- a class $\text{Mor}(\mathcal{C})$ of *morphisms*;
- two functions $\text{dom}, \text{cod} : \text{Mor}(\mathcal{C}) \rightarrow \text{Obj}(\mathcal{C})$ called *domain* and *codomain*, respectively (we write $f : x \rightarrow y$ to say $\text{dom } f = x$ and $\text{cod } f = y$);
- a function $1_\bullet : \text{Obj}(\mathcal{C}) \rightarrow \text{Mor}(\mathcal{C})$ called *identity*;
- a function
$$\circ : \{(g, f) \in \text{Mor}(\mathcal{C}) \times \text{Mor}(\mathcal{C}) \mid \text{dom } g = \text{cod } f\} \rightarrow \text{Mor}(\mathcal{C})$$
called *composition* (we write $g \circ f$ or gf for $\circ(g, f)$);

such that the following axioms are satisfied:

1. $1_x : x \rightarrow x$;
2. composition is associative;
3. the identity 1_x is a unit for composition on both sides.

We write $\mathcal{C}(x, y)$ for the set of morphisms $x \rightarrow y$. We write $x \in \mathcal{C}$ to denote that x is an object of $\mathcal{C}$. We take the fairly common convention that all categories are locally small in these notes. We write $\text{End}_{\mathcal{C}}(x) := \mathcal{C}(x, x)$. We often abuse notation and write 1 for the identity on some object when specifying the object in the subscript becomes inconvenient.

Example 1.1.2. Common categories include

- the category Set of sets and functions;
- the category Grp of groups and group homomorphisms;
- the category Ring of (unital) rings and (unital) ring homomorphisms;
- the category $R\text{-Mod}$ of R -modules and R -linear maps;

- the category $\mathcal{Ch}(\mathcal{A})$ of chain complexes on an abelian category $\mathcal{A}$;
- the category $\mathcal{Top}$ of topological spaces and continuous maps;
- the category $\mathcal{Lin}$ of linearly ordered sets and monotonic maps;
- the category $\mathbf{\Delta}$ of finite nonempty ordinals and monotonic maps;
- the category $\mathcal{Cat}$ of small categories;
- the category $\mathcal{Mod}$ of model categories;
- the category $\mathcal{V}\text{-Cat}$ of small $\mathcal{V}$ -categories for $\mathcal{V}$ some cosmos;
- the category $\mathbf{sSet} = \mathbf{Set}^{\mathbf{\Delta}^{op}}$ of simplicial sets;

and many more.

Given a poset $(S, \leq)$, we define the poset category by saying there is a unique arrow $x \rightarrow y$ whenever $x \leq y$, and no arrow otherwise. We call such a category a *poset category*. All linearly ordered sets are assumed to be poset categories unless specified otherwise. Of particular interest are finite ordinals $n = [n - 1] = \{0, \dots, n - 1\}$ ordered by $\leq$.

Lecture 2

Thu 04 Sep 2025 15:30

New OH: thursday OH 11-12

Chapter 2

Groups

A group is a groupoid with one object.

Lecture 3

Mon 20 Oct 2025 12:21

Lecture 4: Field extensions

Tue 18 Nov 2025 15:36

Theorem 2.0.1. *Let k a field and $f \in k[x]$ irreducible. Then $K := k[x]/(f(x))$ is a field where f has a root. For any field F containing roots of f , there is a homomorphism $K \hookrightarrow F$. All ring homomorphisms are injective.*

Definition 2.0.2. If k, F are fields, then F is a *field extension* of k if $k \subseteq F$. We write F/k . The *degree* of the extension is $\dim_k F$, and is denoted $[F : k]$.

Definition 2.0.3. Let F/k be a field extension with $\alpha \in F$. Then $k(\alpha)$ is the smallest subfield of F containing α and k . It is constructed as

$$k(\alpha) = \bigcap_{k \subseteq F' \subseteq F, \alpha \in F'} F'.$$

Definition 2.0.4. A field extension F/k is *simple* if $F = k(\alpha)$ for some $\alpha \in F$.

Proposition 2.0.5. *Let $k(\alpha)/k$ be simple. Consider the evaluation map $\varepsilon : k[t] \rightarrow k(\alpha)$ given by $t \mapsto \alpha$. Then ε is injective if and only if $k(\alpha)/k$ is an infinite extension. In this case, $k(\alpha)$ is isomorphic to $\text{Frac}(k[t])$, which is the field of rational functions on t .*

If ε is not injective, then there is a monic nonconstant $p(t) \in k[t]$ of degree $n = [k(\alpha) : k]$ such that

$$k(\alpha) \cong \frac{k[t]}{(p(t))}$$

given by $\alpha \mapsto [t]$.